Vasavi Krishna

AI/ML Engineer
+1 (346) 371-3732 | chvasavikrishna@outlook.com | Richardson, TX | LinkedIn

Summary

AI/ML Engineer with 5+ years of experience spanning enterprise networking, security intelligence, deep learning, automation systems, and ML infrastructure. Currently building AI-native infrastructure solutions, including agentic RAG-based assistants, predictive maintenance models, anomaly detection systems, and scalable deployment frameworks. Experienced in LLM fine-tuning using LoRA/QLoRA, distributed training concepts, GPU-enabled workloads, and production ML systems using Docker, Kubernetes, and CI/CD pipelines. Strong foundation in data engineering and DevOps practices, enabling end-to-end ownership from data pipelines and model development to release automation and performance monitoring. Adept at translating complex telemetry and operational knowledge into reliable, high-performance AI solutions.

Technical Skills

Programming & Querying: Python, SQL, R
Machine Learning & Artificial Intelligence: Scikit-learn, TensorFlow, Keras, XGBoost, LightGBM, Random Forest, Logistic Regression, Support Vector Machines (SVM), Deep Learning (CNN, RNN, LSTM, GRU), Natural Language Processing (NLP), Feature Engineering, Hyperparameter Tuning, Model Explainability (SHAP, LIME), Model Fine-Tuning (LoRA, QLoRA), Reinforcement Learning, RLHF (Reinforcement Learning from Human Feedback)
Computer Vision: OpenCV, Object Detection (YOLO, Ultralytics)
Generative AI & LLM Frameworks: Hugging Face Transformers (BERT, GPT, T5), LangChain, LlamaIndex, LangGraph, Prompt Engineering, Retrieval-Augmented Generation (RAG), AutoGen, Haystack, LLM Agents / Agentic Workflows, LLM Evaluation & Benchmarking, Evaluation Dataset Creation, LLM Performance Metrics
Data Engineering & Real-Time Processing: Apache Kafka, Apache Spark, PySpark, Apache Airflow, MLflow
APIs & Backend Development: FastAPI, Flask, RESTful APIs
Data Analytics & Statistics: Pandas, NumPy, SciPy, Statistical Analysis, A/B Testing, Time Series Analysis, Predictive Analytics
Data Visualization & Business Intelligence: Tableau, Power BI, Matplotlib, Seaborn, Plotly
Databases & Vector Stores: MySQL, PostgreSQL, Oracle, MongoDB, Data Warehousing Concepts, FAISS, Pinecone
Cloud Platforms & Deployment: AWS (EC2, S3, SageMaker, Lambda), Azure (Azure ML, Azure Data Factory, Azure Blob Storage), GCP (BigQuery, Vertex AI, Cloud Storage), Docker, Kubernetes, Streamlit, Cloud ML Pipelines
Monitoring & Production Readiness: Model Performance Monitoring, Data Drift Detection, Concept Drift Detection, Error Handling
Version Control & CI/CD: Git, GitHub, GitHub Actions, GitLab, GitLab CI/CD, Bitbucket

Experience

Cisco — Richardson, TX
AI/ML Engineer — *Jun 2024 – Present*

- Designed and implemented agentic network assistants using LangChain and RAG pipelines with FAISS and Pinecone, enabling automated querying of technical documentation and reducing Level-1 resolution time by approximately 40%.
- Built predictive maintenance models using LSTM, XGBoost, and scikit-learn on time-series telemetry data, translating outputs into actionable insights through feature engineering and statistical analysis techniques.
- Developed anomaly detection systems using deep learning models such as autoencoders and TensorFlow in GPU-enabled environments, improving real-time threat detection and operational visibility across network systems.
- Deployed containerized machine learning services using Docker and Kubernetes, integrating CI/CD pipelines with GitHub Actions for automated testing, version control, and streamlined model deployment workflows.
- Implemented model performance monitoring using MLflow, along with automated validation workflows, data drift detection, and concept drift tracking to maintain consistent reliability in production environments.
- Collaborated with platform teams using Python and SQL on Linux-based environments, optimizing resource utilization, improving data workflows, and supporting scalable inference deployments across distributed systems.

Tungsten Automation — Remote, USA
Automation Engineer — *Jul 2023 – Apr 2024*

- Built Python-based workflow automation tools and shell scripts to manage recurring data and infrastructure tasks, reducing manual effort and improving reliability.
- Assisted in configuring CI/CD workflows with Git-based version control, automated testing, and deployment pipelines to enhance release consistency.
- Monitored execution logs, optimized pipeline performance, and improved system stability across distributed environments.
- Integrated internal systems via REST APIs while maintaining secure and structured deployment practices.

Sigma Infosolutions Ltd. Hyderabad, India
Data Engineer *May 2020 – Dec 2022*

- Developed advanced SQL queries and Python scripts for exploratory data analysis and predictive modeling fueling enterprise analytics initiatives that helped scale ML pilots into production solutions.
- Designed ETL workflows and data preprocessing pipelines using Azure Synapse and DBT accelerating cloud data modernization for legacy system migration, cutting migration time by up to 50% and enabling faster analytics-ready data.
- Performed EDA on large financial and e-commerce datasets by visualizing patterns and trends using seaborn and Plotly interactive dashboards revealing inefficiencies that drove workflow automation, achieving 70% manual effort reduction.
- Built interactive Power BI and Tableau dashboards with DAX, KPIs, and clear data storytelling, tracking A/B testing outcomes and operational performance insights.

Projects

Agentic Network Troubleshooting Intelligence System (Cisco)

- Architected a production-grade agentic troubleshooting system using transformer-based embeddings, semantic vector indexing in Pinecone, and LangChain-orchestrated RAG pipelines with dynamic chunking, top-k retrieval optimization, contextual memory handling, and response grounding controls to deliver deterministic, low-latency root-cause recommendations across distributed enterprise network environments.

Network Traffic Anomaly Detection Using Hybrid LSTM-Attention Architecture

- Built a hybrid LSTM-Attention deep learning model for multivariate time-series network telemetry, incorporating feature normalization, sliding-window sequence modeling, and adaptive thresholding to detect temporal anomalies, improving early outage detection accuracy while minimizing false positives in real-time monitoring systems.

Domain-Adaptive LLM Fine-Tuning Framework

- Fine-tuned a compact transformer-based LLM using LoRA and QLoRA with 4-bit quantization, training on curated network incident tickets and resolution logs to adapt domain-specific terminology while significantly reducing GPU memory usage and avoiding full-parameter model retraining overhead.

Publications

Krishna, V. (Co-Author), et al.

Certifications

Microsoft Certified: Power BI Data Analyst Associate (PL-300)

IBM Data Analyst Professional Certificate

Education

Central Michigan University Mt. Pleasant, Michigan
Master's, Computer Science *Jan 2023 – May 2024*
Malla Reddy Institute of Technology Hyderabad, India
Bachelor's, Computer Science *Aug 2017 – Jul 2021*